

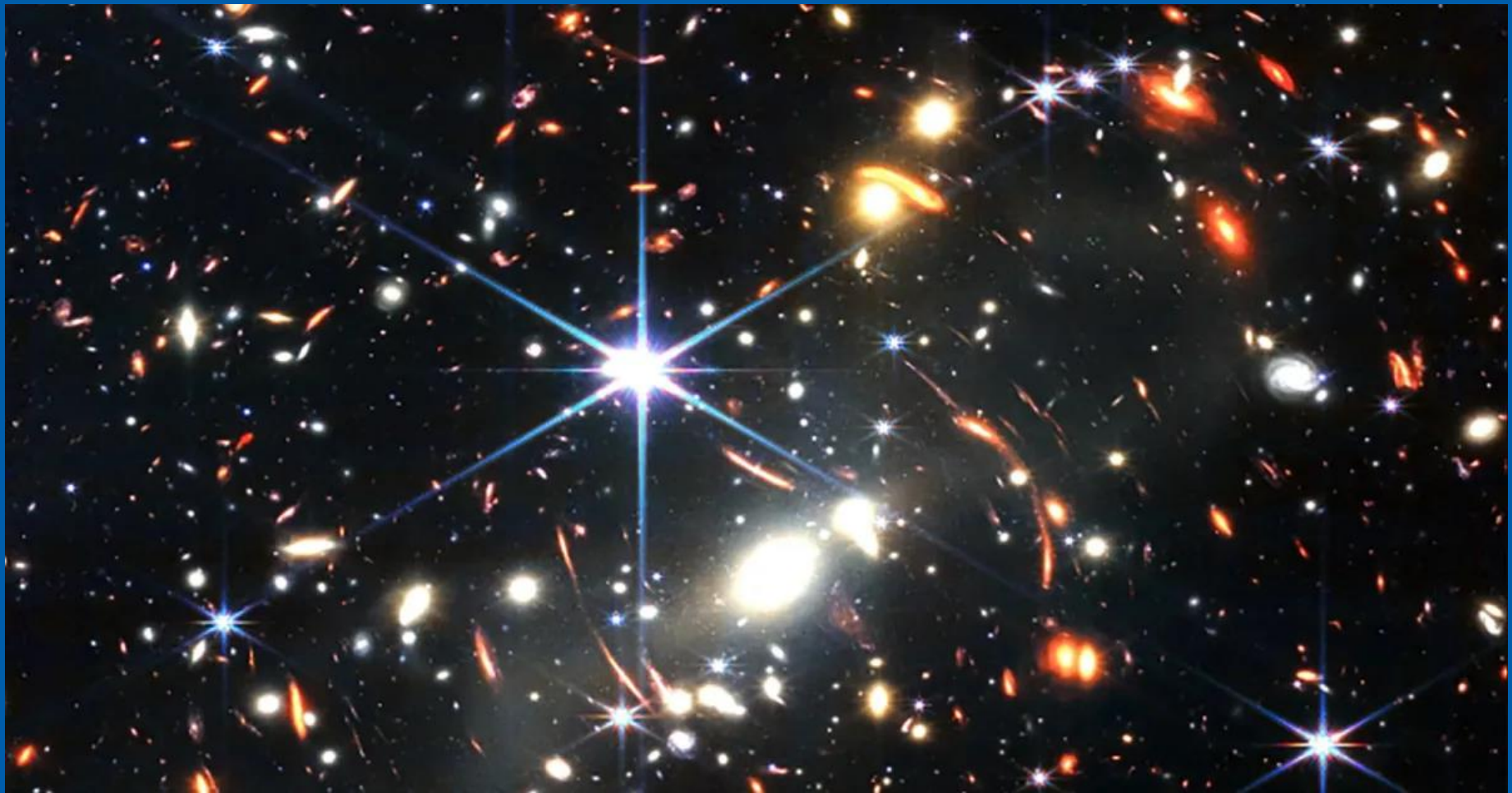
Astronomy 102 - Lab 12

Hubble's Law and the Expanding Universe

Class Begins 12:05

Check out: <https://www.webbcompare.com/>

Matthew Prest

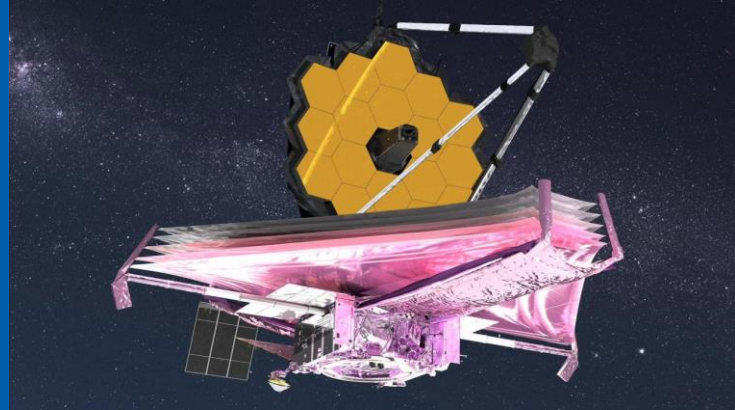


James Webb Deep Field

The latest in Astronomy

Galaxy Rotation

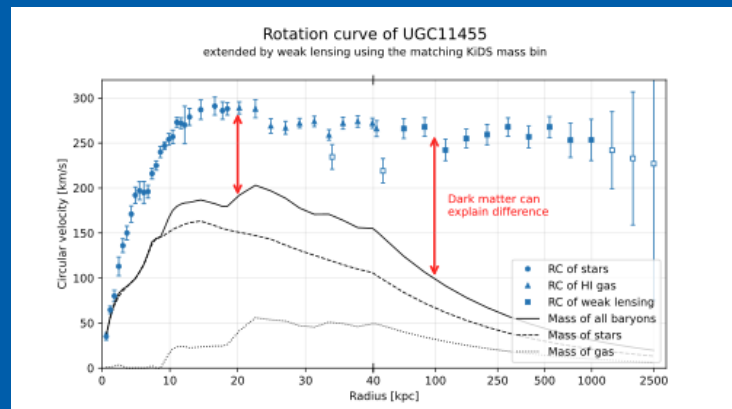
- Astronomers using the James Webb Space Telescope have made a surprising discovery about a galaxy long, long ago and far, far away: It isn't rotating.
- Some galaxies merge with each other multiple times and their combined rotations added to or partly canceled each other.
- This usually takes a long time, so finding a young galaxy with little rotation is unusual
- Galaxy rotation is a key piece of evidence for dark matter!
- https://upload.wikimedia.org/wikipedia/commons/transcoded/3/33/Galaxy_rotation_under_the_influence_of_dark_matter.ogv/Galaxy_rotation_under_the_influence_of_dark_matter.ogv.240p.vp9.webm



The latest in Astronomy

Galaxy Rotation

- Suspected Dark Matter “Halos” around the edges of galaxies
- The edges of galaxies should rotate more slowly, instead they seem to have a roughly constant rate of rotation!
- One of the biggest open problems in astronomy!



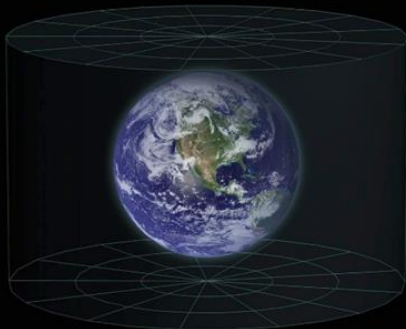
Question

Can you compare the height of two distant people?

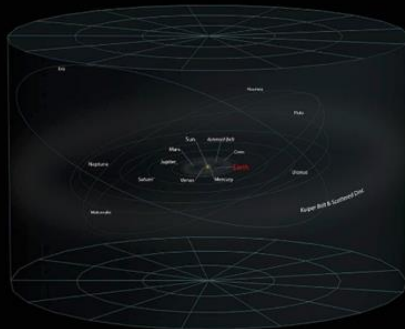
- How?

Remember our first lab?

Earth



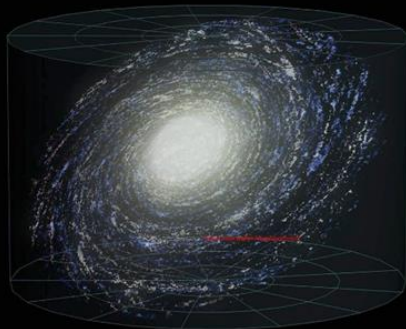
Solar System



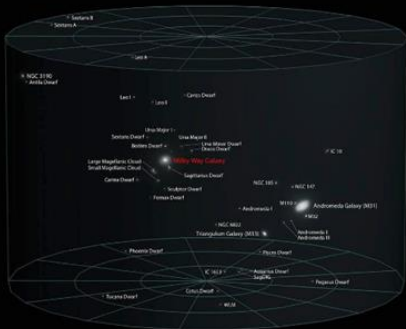
Solar Interstellar Neighborhood



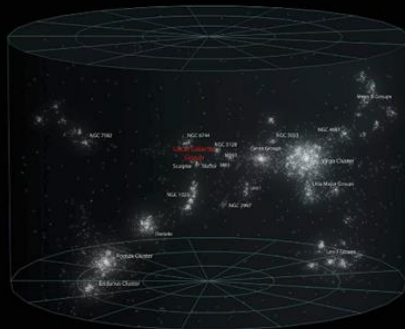
Milky Way Galaxy



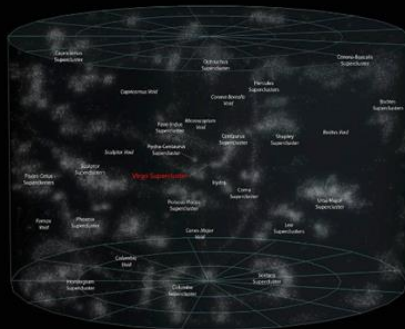
Local Galactic Group



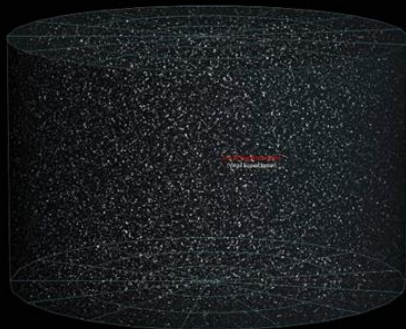
Virgo Supercluster



Local Superclusters

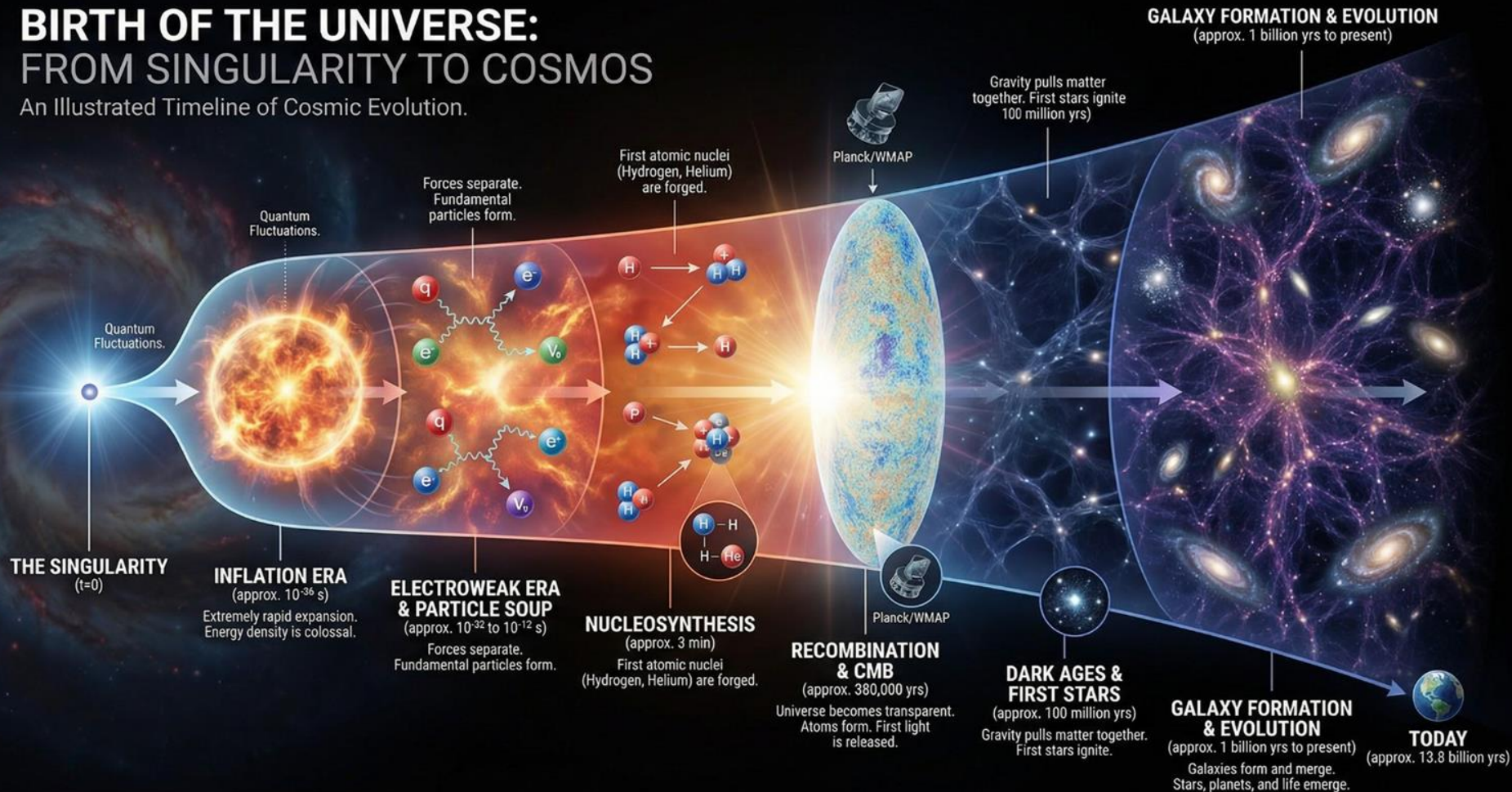


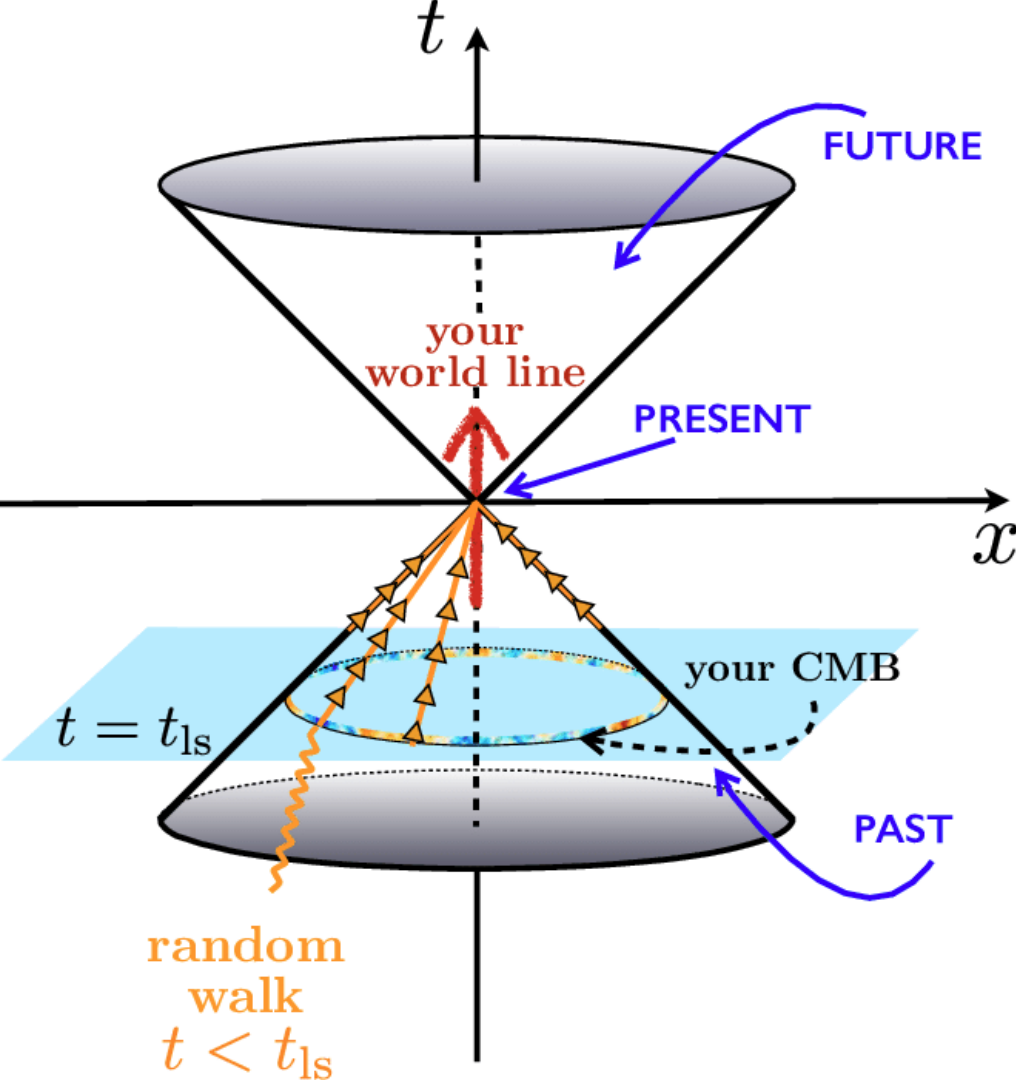
Observable Universe



BIRTH OF THE UNIVERSE: FROM SINGULARITY TO COSMOS

An Illustrated Timeline of Cosmic Evolution.

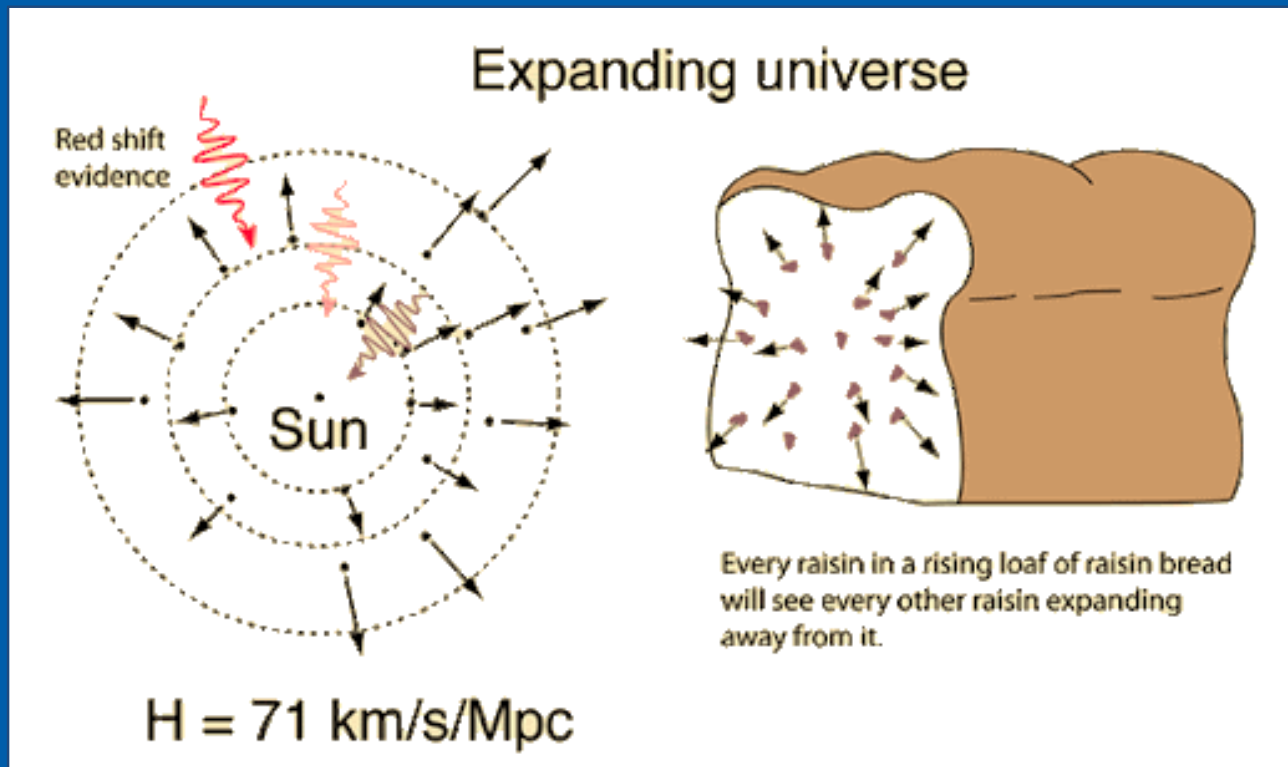




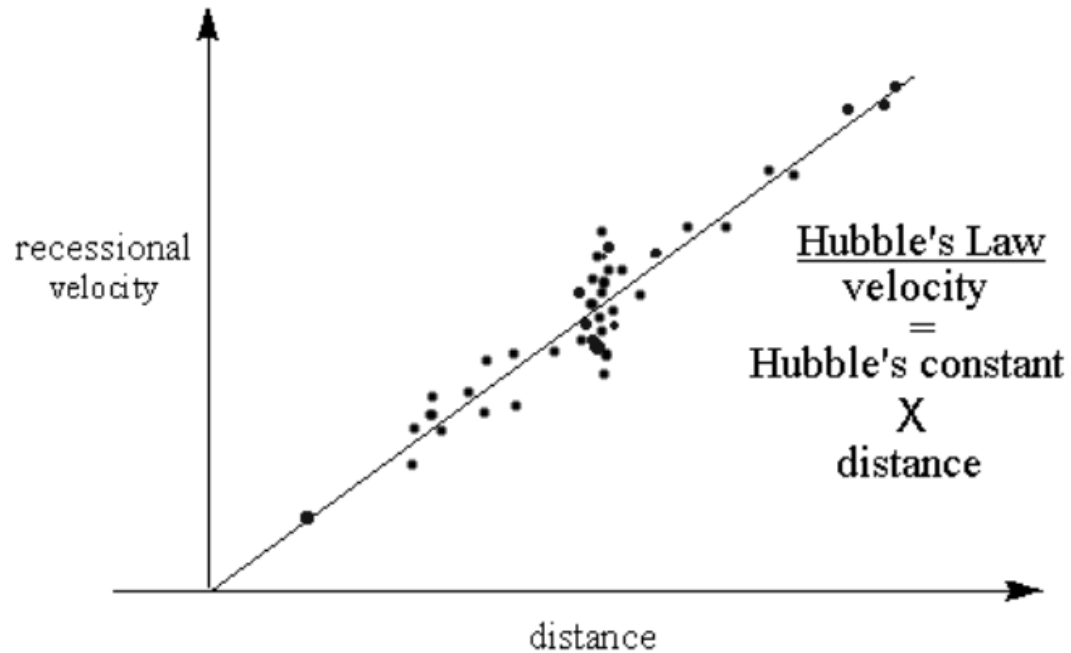
Special relativity tells us that space and time are inseparably connected

- One object's past is not the same as another's!
- The present "time slice" depends on relative velocity
- When we think about expansion, it is expansion with time that we observe

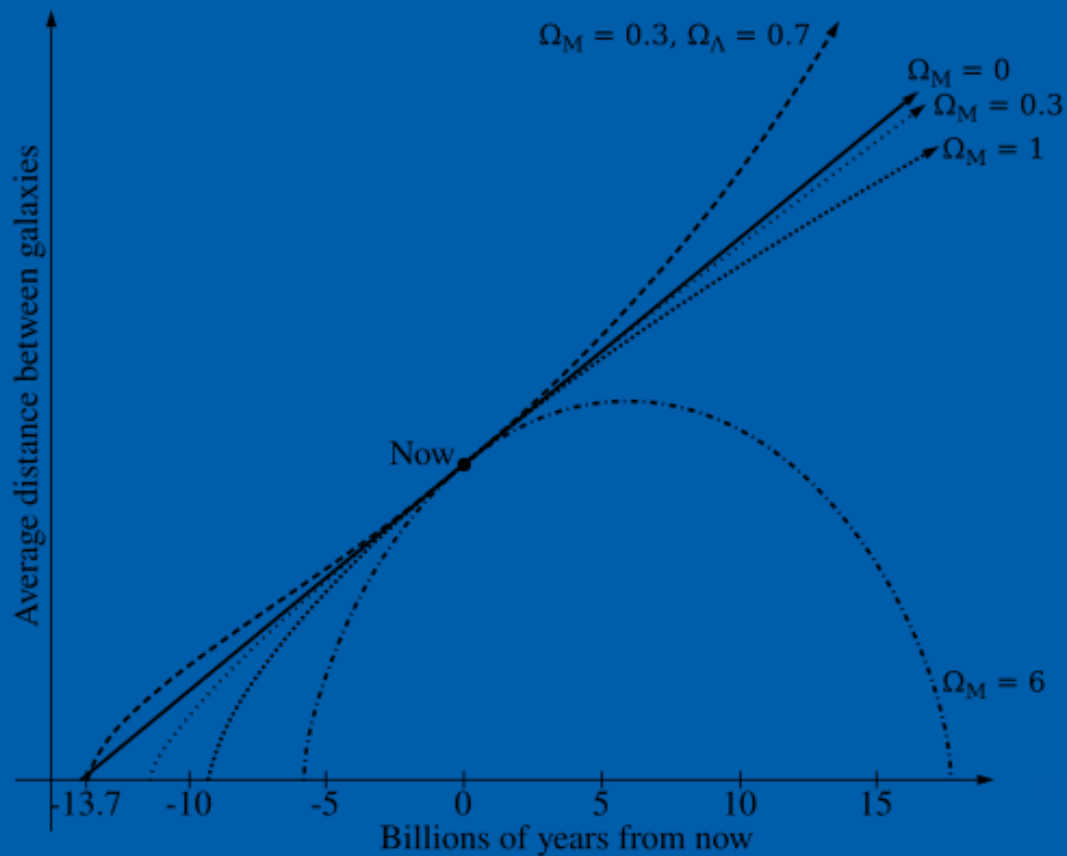
Hubble Lemaître law



Hubble Lemaître law



Ultimate Fate of the Universe



Makeup In-person Lab

1. In person, May 14th, 12pm-2:50pm Hunter CN115 (usual room)
1. Can only do 1 lab!
 - Either a lab you missed or a lab you want to improve your grade
 - Last in-person lab grades will be posted later tonight
1. All In-Person lab grades are available on brightspace
1. No print outs needed, I will prepare copies for each experiment.

Reminders

1. Complete the Tutorial first!

- This is best done now in class where I am available to assist

1. Attempt the essay questions last!

- These should be based on what you have covered in the tutorial.

1. Previous week's lab report due at the beginning of the next lab!

- Final grades are based on best 6 online labs plus each of the 4 in-class labs

1. Grades will be posted to Brightspace

- Check now that you have access to our course!
- Grade feedback will also be provided on Pearson, please check as this is the best way to improve your future grades!

1. Office hours

- Office hours are available on Zoom Mondays 4-6pm (same zoom link as class) as well as by appointment either in-person on or zoom. Come say hi, I'm here to help!

Any and all questions are welcome!